GeoCNN-10 (PlonkNet): Country-Level Geolocation of Street View Images with a Pretrained CNN

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***Abstract*—This project studies whether a standard pretrained convolutional neural network can identify the country of a single Street View image, as players do in the game GeoGuessr. The data come from the Kaggle dataset GeoLocation – Geoguessr Images (50K). The project starts with a data diagnosis covering file integrity, exact and near-duplicates, image resolution and class imbalance produced a clean set of 13,527 images from 10 countries, divided into a group that players consider easy to distinguish and a group they consider hard, according to the Plonk It guide.**

A ResNet50 pretrained on ImageNet was kept frozen, and only a new linear head was trained on cached features. On the test set, the baseline reaches a top-1 accuracy of 62.6%, a top-5 accuracy of 94.7% and a macro F1 of 0.585, far above chance (10% and 50%). The easy group obtains a clearly higher F1 than the hard group (0.685 vs. 0.485), France attracts most of the errors from the other European countries, Italy is the weakest class, and the model turns out to be underconfident rather than overconfident. [ToDo: poner mejoras ajuste del modelo y su resultado]

***Index Terms*—Deep Learning, Transfer Learning, Image Geolocation, ResNet50, Model Calibration, GeoGuessr**

I. Introduction

GeoGuessr is a game in which players see a Street View image and must guess where it was taken. In the *No Move, Pan, Zoom* (NMPZ) mode, the player gets a single fixed view and cannot move, rotate or zoom. This project studies the same setting for a neural network: a single frame, at a reduced resolution (224 × 512 pixels), classified in one of 10 countries.

From my own experience as a casual GeoGuessr player, two of the first things I look for are text in the local language and the position of the sun, which tells whether the location is in the Northern or Southern Hemisphere. In NMPZ mode, however, those clues are often not in view, and there is no way to move and look for them. What motivated this project was noticing my own thought process in those cases: I rely on mental models and visual archetypes, such as what European architecture looks like, Japanese roofs and compact cars, tuk-tuks in Thailand, pickup trucks in the United States, or lush green vegetation in Brazil. This made me wonder whether a model could also abstract high-level visual features and predict a country in a similar way, without depending on fine-grained textual clues.

Experienced players, on the other hand, rely on very specific clues, such as the shape and color of bollards and utility poles, license plates, road markings and the language of the signs, which guides like Plonk It [1] document country by country. Interestingly, they also memorize "meta" clues, such as the quality of the camera and the type of coverage of each region, as well as the soil and vegetation. At 224 × 512 pixels, a CNN cannot read signs or see small details, but it can be very good at capturing general patterns, so it is closer to the casual player than to the expert. And it may also pick up some of the same meta clues the experts use, such as the camera quality or the lighting. The question is whether these general impressions are enough to tell countries apart, and which of them the model actually uses.

Research question: can a standard pretrained CNN, without access to readable text or a specialized training on objects in images, difference countries from the scenery alone, and does it fail mainly where human players also find it hard?

The objectives are:

1) To build an end-to-end pipeline (data diagnosis, preparation, training, evaluation, and inference) for 10 countries.
2) To compare a group of countries that players consider easy to distinguish with a group they consider hard.
3) To evaluate the calibration of the model and the effect of class imbalance.
4) To evaluate and rate the performance and accuracy of the model baseline version and final version.

Evaluation metrics. To assess the model performance, generalization, and calibration, three primary metrics are employed:

- **Macro F1-Score:** Assigns equal weight to all 10 classes to prevent class imbalance from biasing the evaluation. It combines Precision (minimizing false positives by ensuring predicted countries are correct) and Recall (minimizing false negatives by capturing all actual country images). Averaging these per-class F1 scores equitably ensures that minority countries carry the exact same importance as majority classes in the final evaluation.
- **Top-1 and Top-5 Accuracy:** Operating on the model's output probability distribution, **Top-1 Accuracy** checks if the single highest-probability prediction matches the ground truth. **Top-5 Accuracy** considers a prediction correct if the true country lies within the model's top 5 highest-probability candidates a crucial metric in geographic tasks for evaluating whether the model groups visually similar regions together or places the correct country within its top candidate pool.

- **Expected Calibration Error (ECE):** Quantifies the reliability of the model's confidence scores. If the network predicts a group of images with 90% confidence, we expect it to be correct 90% of the time. ECE measures the weighted gap between these predicted confidences and empirical accuracies across confidence bins, allowing us to detect whether the model is overconfident (highly certain but incorrect) or underconfident.

Success criteria. The main metric is the macro F1 score, however the model will be considered successful if

- It performs clearly above chance (10% top-1 and 50% top-5 for 10 classes).
- The easy group obtains a higher F1 than the hard group, as expected from the player guides.
- The adjustments after the baseline improve the macro F1 and the calibration error.

Related work. DeepGeo [2] trained a ResNet on Street View images of the 50 US states. PIGEON [3], trained on GeoGuessr data, won six out of six games against one of the world's best professional players. G^3 [4] uses a GeoGuessr guidebook written by humans to improve country prediction, and a Stanford CS229 project [5] applied transfer learning with ResNet-50 to the same Kaggle dataset used here.

Contribution. This project does not aim at a new state of the art or is not expected to over-perform the previous mentioned models. Its success is measured in a controlled study: easy and hard groups defined from a player guide, near-duplicate removal to avoid leakage between splits, a calibration analysis, interpretability with Grad-CAM [6], and a comparison with a human player.

The work is organized as follows:

- Section II describes the data.
- Section III the diagnosis of the dataset.
- Section IV the model and its justification.
- Section V the baseline results.
- Section VI the adjustments.
- Section VII the inference notebook, and the last sections the limitations and conclusions.

II. Data

A. Source

The data come from the Kaggle dataset *GeoLocation – Geoguessr Images (50K)* [7]: 49,997 GeoGuessr screenshots organized in 124 folders, one per country or territory, where the folder name is the label. Almost all the images measure 1536×662 pixels and less than 1% is in another format resolution. According to the dataset page, the screenshots were collected with the Python libraries Selenium and geopy. The license is not specified by the author (it is listed as "Unknown" on Kaggle).

B. Country selection and groups

Countries with few images were excluded, and 10 countries were selected among the 24 with the most images. Following the Plonk It guide [1], five countries were placed in a group that players consider easy to distinguish and five in a group they consider hard. Among the candidates, the countries with the most images were chosen (Table I).

TABLE I
SELECTED COUNTRIES, FINAL NUMBER OF IMAGES, GROUP, AND PRIMARY VISUAL CLUES.

Country	Images	Key Clue (Plonk It)	Group
United States	2,000	Double-yellow centerlines	Easy
Japan	2,000	Kanji/Kana poles, low camera	Easy
Brazil	2,000	Red soil, Latin script	Easy
South Africa	1,177	Yellow outer shoulders, chevrons	Easy
Thailand	941	Dense cables, Thai script	Easy
France	2,000	Rural pine forests, white/red bollards	Hard
Spain	1,071	Mediterranean architecture	Hard
Poland	857	Red/white U-12a bollards	Hard
Italy	787	Dual blue strip front plates	Hard
Germany	694	Gen 3 blur, typography	Hard

Dataset Group Analysis and Clue Distribution:

- **The "Easy" Group (Macro Clues):** These countries are characterized by striking structural and environmental features. The United States features clear double-yellow centerlines; Japan relies on low-camera heights and dense character blocks (Kanji/Kana) on utility poles. Brazil is identifiable by red soil and Latin script, South Africa by yellow outer shoulder lines and high-contrast chevron signs, and Thailand by left-hand driving combined with dense cable bundles and complex Thai script textures.
- **The "Hard" Group (Micro Clues & Pixelation):** Comprising a European cluster under the Vienna Convention, these countries share uniform macro-scenery, road layouts, and temperate vegetation. Because macro features overlap heavily, differentiation relies on fragile *micro-clues* (such as bollard color caps or license plate geometry). At the input resolution of 224×512 pixels, these fine details degrade and pixelate, forcing the model to struggle with regional nuances (e.g., distinguishing French vs. Polish bollards or Spanish vs. Italian plates).

III. Data Diagnosis

To make a diagnois tests were made trough a mini ETL and a mini EDA. Each test answers one question: is there a problem in the data that could distort the model's results or its confidence? Table II summarizes the tests, their results and the actions taken.

Fig. 1 shows that the only duplicate founded was this one and was not in the contenders but is evidence of a successful code hence supporting that none of the trained data has a duplicate and producing data leakage, the image is a guaranty. Fig. 2 shows the near duplicate sampled and it would be a real threat if it wasn't attended producing data leakage. Fig. 3 shows the imbalance among the 10 selected countries before the cap: the United States alone has 12,014 images, more than three times as many as Japan, the second largest. Fig. 4 shows 20 random images of the United States and France: in rural

TABLE II
DIAGNOSIS TESTS, RESULTS AND ACTIONS

Test	Result	Risk for the model	Action
File types and integrity	All files are valid JPG images	—	None
Color mode	All images are RGB	—	None
Exact duplicates (MD5)	1 duplicate pair (Ghana)	Inflated test results	Redundant copy flagged and excluded
Image resolution	55 images (0.11%) at 1030 × 703; the rest at 1536 × 662	Image shape as a shortcut	Spread across 28 countries (at most 0.29% of a selected country): no action
Class imbalance	From 12,014 images (United States) to 1 image (several territories)	Over-prediction and overconfidence for large classes	Cap of 2,000 images per country, random with a fixed seed
Near-duplicates (pHash)	94 images (0.69%) in 47 pairs	Leakage between splits, inflated accuracy	22 replaced with unused images, 25 removed
Visual inspection (200 images)	No label errors, non-street images, game UI elements, watermarks or black borders. Rural scenes of the United States and France look alike; the main difference is the road markings	Cross-group confusion	Both countries kept on purpose (hypothesis 5)

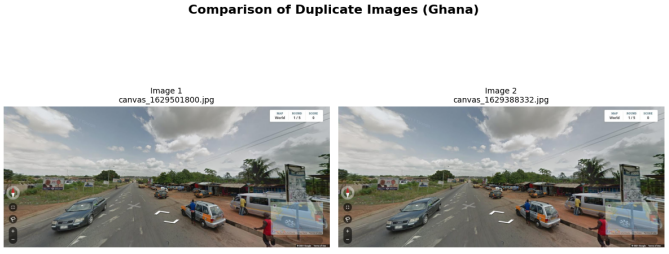


Fig. 1. Duplicate found.



Fig. 2. Example of near duplicates.

scenes without buildings, both share similar vegetation and lighting, and the main difference is the road markings (double yellow center lines in the United States, white lines in France).

A. Final dataset and split

The final dataset has 13,527 images. It was split into training (9,661), validation (1,933) and test (1,933) sets, about 71/14/14%, stratified by country. The test set is used only once, to report the final result. A single split was used instead of k-fold cross-validation, because training the network several times did not fit the project time.

B. Hypotheses

Before training, the diagnosis and the Plonk It guide led to five hypotheses:

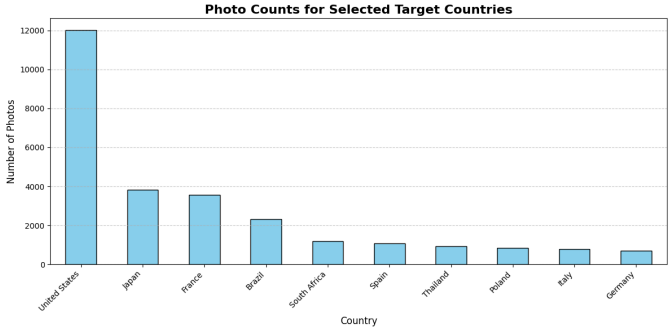


Fig. 3. Number of images of the 10 selected countries before the cap of 2,000 images per country.

1) **Imbalance and calibration:** the countries with the most images (United States, Japan, Brazil and France) will receive more wrong predictions, and the model will be overconfident in them, because models develop a bias towards majority classes during training.
2) **Easy group:** Brazil and Thailand will be confused with each other, because both have dense tropical vegetation and rural unpaved roads; Japan and the United States will be confused in suburban streets and temperate forests without text. South Africa will be the best distinguished, as the only African country in the sample, with semi-arid landscapes and yellow shoulder lines.
3) **Hard group:** France and Poland will be the hardest to identify, because both share flat rural landscapes, fields and European road signs; Italy will be the easiest of the group thanks to its license plates and bollards. These are clues for human players, and whether they survive at this resolution is part of what the experiment tests.
4) **Text vs. scenery:** at this resolution most signs are unreadable, so the model will rely on scenery (vegetation, road markings, buildings and colors) rather than on text.
5) **Cross-group confusion:** the United States and France



Fig. 4. Random samples of 20 images: United States (left) and France (right).

TABLE III
HOW THE DIAGNOSIS SHAPED THE MODEL DESIGN

Diagnosis finding	Decision
About 9,700 training images: too few to train a deep network from scratch	Transfer learning: ResNet50 [8] pretrained on ImageNet [9]
Panoramic images (ratio ≈ 2.3)	Input of 224×512 instead of a square crop
Class imbalance after the cap (2,000 vs. 694 images)	Macro F1 as the selection metric; no class weights in the baseline, to observe the effect
Near-duplicates	Removed before the split
Driving side and text orientation are real clues	No horizontal flips (no augmentation in the baseline)

TABLE IV
TRAINING CONFIGURATION OF THE BASELINE

Setting	Value
Input	224×512, ImageNet normalization
Loss	Cross-entropy, no class weights
Optimizer	AdamW [11], learning rate 10^{-3}, weight decay 10^{-4}
Batch size / epochs	64 / 10, best epoch selected by validation macro F1
Seed	42

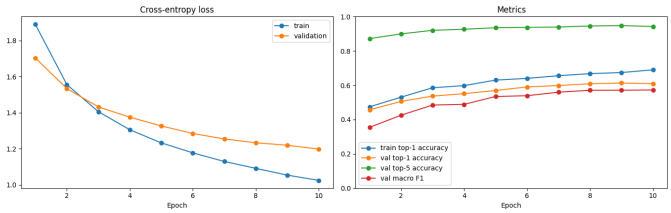


Fig. 5. Loss and metrics per epoch of the baseline.

will be confused even though they belong to different groups, because the visual grids show similar green rural landscapes, roads and lighting.

IV. MODEL

A. Choice of model based on the diagnosis

Table III links each finding of the diagnosis to a design decision.

B. Architecture and training

The model was implemented in PyTorch [10]. The pre-trained ResNet50 is frozen, and its original last layer (1,000 ImageNet classes) is replaced with a new linear layer of 10 outputs, initialized at random: 20,490 trainable parameters out of 23.5 million. Because the backbone is frozen and there is no augmentation, every image always produces the same 2,048 features, so they were computed once and cached (linear probing). This gives the same result as passing the images through the network in every epoch, and it made training possible without a GPU. Table IV lists the training configuration.

C. Advantages and disadvantages

Advantages: it needs little data; it trains fast, even on a CPU with cached features; it is a standard and well-studied architecture; and it is compatible with Grad-CAM. **Disadvantages:** the ImageNet features were learned on objects, not on landscapes; frozen features cannot adapt to street scenes; the input resolution erases small clues such as signs and plates; the model weighs about 100 MB, which is heavy for deployment; and it may rely on superficial cues such as color or lighting. [ToDo: toca investigar mas de la propuesta]

V. BASELINE RESULTS

The validation loss was still decreasing at epoch 10 and the best epoch was the last one (Fig. 5), so the head had not converged yet. The gap between training and validation (69% vs. 61% top-1) is a moderate generalization gap, but not harmful overfitting, since the validation loss never went up. Table V shows the results. [ToDo: TEerminar de explciar que no converge para ponerle mas epocas]

On validation, the easy group reaches an average F1 of 0.665 and the hard group 0.480 (0.685 and 0.485 on test). Japan (0.722) and Germany (0.717) obtain the highest F1, and Italy the lowest (0.212): only 14% of its images are

TABLE V
BASELINE RESULTS (ECE: EXPECTED CALIBRATION ERROR [12])

Split	Top-1	Top-5	Macro F1	ECE
Validation	60.9%	94.3%	0.573	0.085
Test	62.6%	94.7%	0.585	0.102
Chance	10%	50%	≈0.10	—

recognized, and the rest go mainly to France (22%), Spain (21%) and Japan (14%). France works as a "sink" for the European errors: 17–22% of the images from Germany, Italy, Poland and Spain are predicted as France. The reliability diagram stays above the diagonal (Fig. 7): the model is right more often than its confidence says, so it is underconfident.

[ToDo: Describir el val y el test]

A. Hypotheses check

Table VI compares the hypotheses with the validation results.

VI. Adjustments

[ToDo: hacer, correr y poner la mejora]

VII. Inference

The prediction notebook loads the trained weights and returns the five most likely countries for any image, both as text in the console and as a chart. A screenshot is first center-cropped to the panoramic ratio used in training. As a qualitative demo, 20 random Street View images (2 per country) were classified: 11 of 20 were correct in the top-1 prediction and all 20 had the right country in the top 5. The errors follow the patterns of the confusion matrix: 4 of the 9 errors went to France, both Italian images were misclassified, and both Spanish images were predicted as South Africa, probably because of their arid landscape. With only 20 images, these numbers are a demonstration, not an evaluation. [ToDo: pomer predictt del notebook 3 pre ajuste y post ajuste]

VIII. Human Comparison

[ToDo: ya no necesario si da tiempo nomas]

IX. Limitations and Future Work

The dataset has no coordinates, so the regional distribution inside each country could not be checked. Only exact perceptual-hash matches were treated as near-duplicates. The second dataset, Streetview by Country, could add images for the smaller countries, as long as it is added to all 10 countries so that the image source does not become a shortcut. Fine-tuning the last block of the ResNet, augmentation, other backbones (EfficientNet, ConvNeXt, ViT) and cross-validation remain as future work. [ToDo: revisar]

X. Conclusion

[ToDo:]

References

[1] "Plonk it: A step-by-step guide to geoguessr," Online guide, accessed November 2026. [Online]. Available: https://www.plonkit.net/guide

[2] S. Suresh, N. Chodosh, and M. Abello, "DeepGeo: Photo localization with deep neural network," 2018. [Online]. Available: https://arxiv.org/abs/1810.03077

[3] L. Haas, M. Skreta, S. Alberti, and C. Finn, "PIGEON: Predicting image geolocations," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[4] G. Luo, G. Biamby, T. Darrell, D. Fried, and A. Rohrbach, "G^3: Geolocation via guidebook grounding," 2022. [Online]. Available: https://arxiv.org/abs/2211.15521

[5] "CNN plays GeoGuessr: Transfer learning on ResNet50," Stanford CS229 final report. [Online]. Available: https://www.finndayton.com/CS229_Final_Report.pdf

[6] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.

[7] ubitquitin, "Geolocation – geoguessr images (50k)," Kaggle dataset, accessed November 2026. [Online]. Available: https://www.kaggle.com/datasets/ubitquitin/geolocation-geoguessr-images-50k

[8] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[9] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009.

[10] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[11] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[12] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2017.

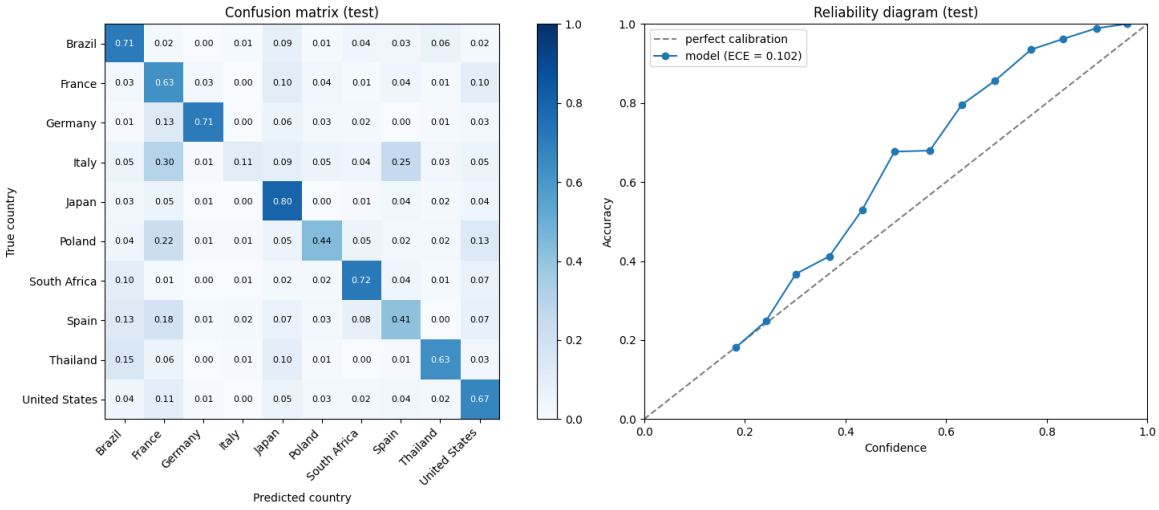


Fig. 6. Confusion matrix (rows: true country, normalized per row) and reliability diagram on the test set.

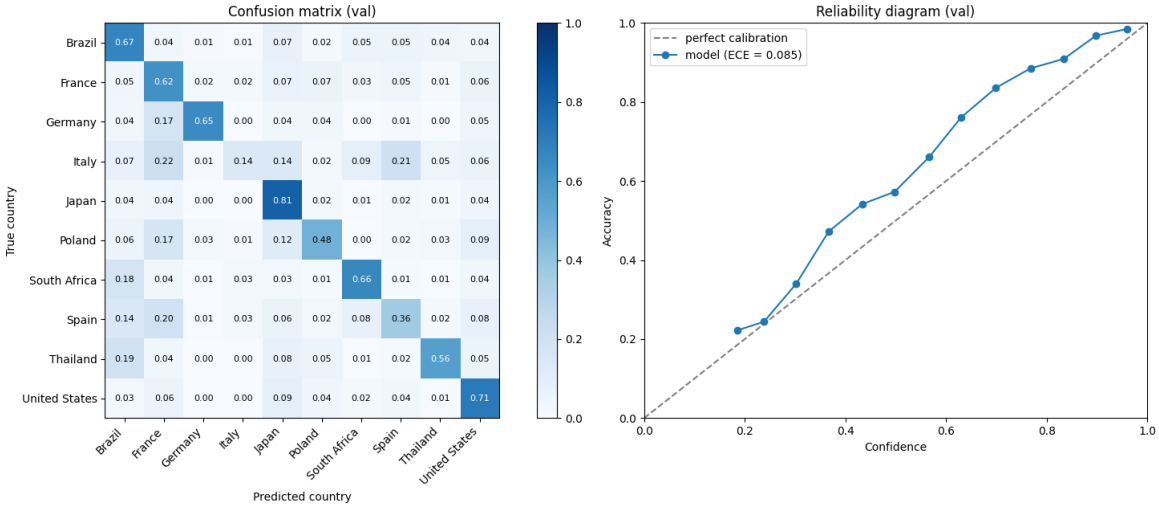


Fig. 7. Confusion matrix (rows: true country, normalized per row) and reliability diagram on the test set.

TABLE VI
HYPOTHESES AGAINST THE BASELINE RESULTS (VALIDATION)

Hypothesis	Evidence	Supported?
1. Imbalance and calibration	Large classes are over-predicted: Japan 18.5% of the predictions vs. 14.8% of the images, France 17.0% vs. 14.9%, Brazil 16.5% vs. 14.7%. But the model is underconfident, not overconfident (ECE 0.085)	Partially
2. Easy group	Easy F1 0.665 vs. hard 0.480. Thailand is predicted as Brazil in 19% of the cases. Japan is the best country (0.722), not South Africa, which goes to Brazil in 18% of the cases	Partially
3. Hard group	Italy is the worst class (F1 0.212), not the easiest of the group; its human clues (plates, bollards) probably do not survive at this resolution. Germany, the smallest class, reaches 0.717	No
4. Text vs. scenery	[ToDo: pendiente: revisarlo con Grad-CAM [6]]	[ToDo:]
5. United States – France	Only 6% of confusion in each direction	Weak

TABLE VII
EXPERIMENT LOG (VALIDATION)

Run	Change	Top-1	Macro F1	ECE
1	Baseline (frozen ResNet50, linear head, 10 epochs)	60.9%	0.573	0.085
2	[ToDo: más épocas]			
3	[ToDo: cabeza MLP]			
4	[ToDo: pesos por clase]			
5	[ToDo: temperature scaling]			